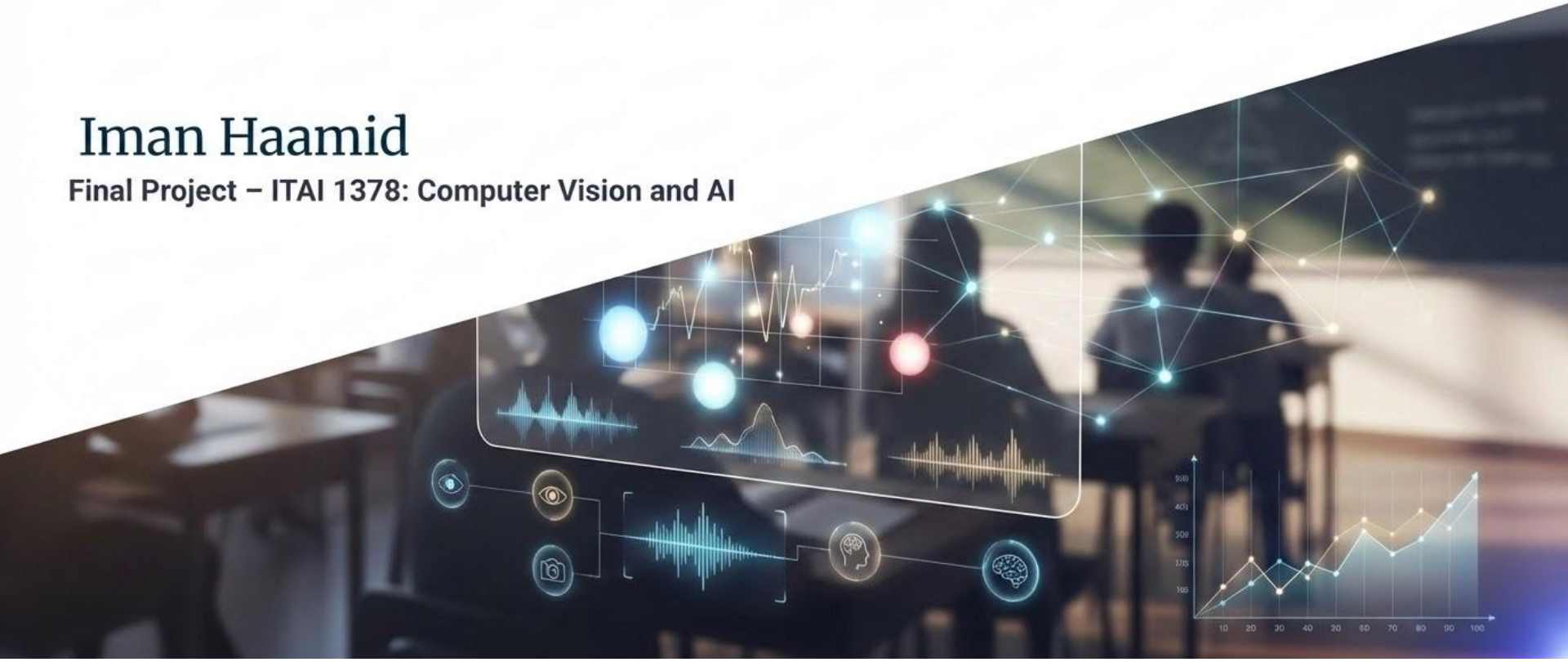


FocusLens

Real-Time Student Engagement and Attention Monitor

Iman Haamid

Final Project – ITAI 1378: Computer Vision and AI



Problem: Lack of Student Engagement



Challenge: Instructors lack immediate feedback on whether students are paying attention or drifting off.

Solution: FocusLens

- Real-time monitoring of student engagement
- Computer vision detects attention and focus
- Provides instructors with immediate valuable feedback



Solutions - FocusLens



What it is?

A computer vision system that monitors student attentiveness via webcam.



Key Function:

Detects attention and focus in real-time.



Goal:

Provide valuable feedback to optimize the learning environment.



Technical Tools and Models



Language: Python (Standard for CV/ML).

Core Libraries:



OpenCV: Image acquisition and processing.



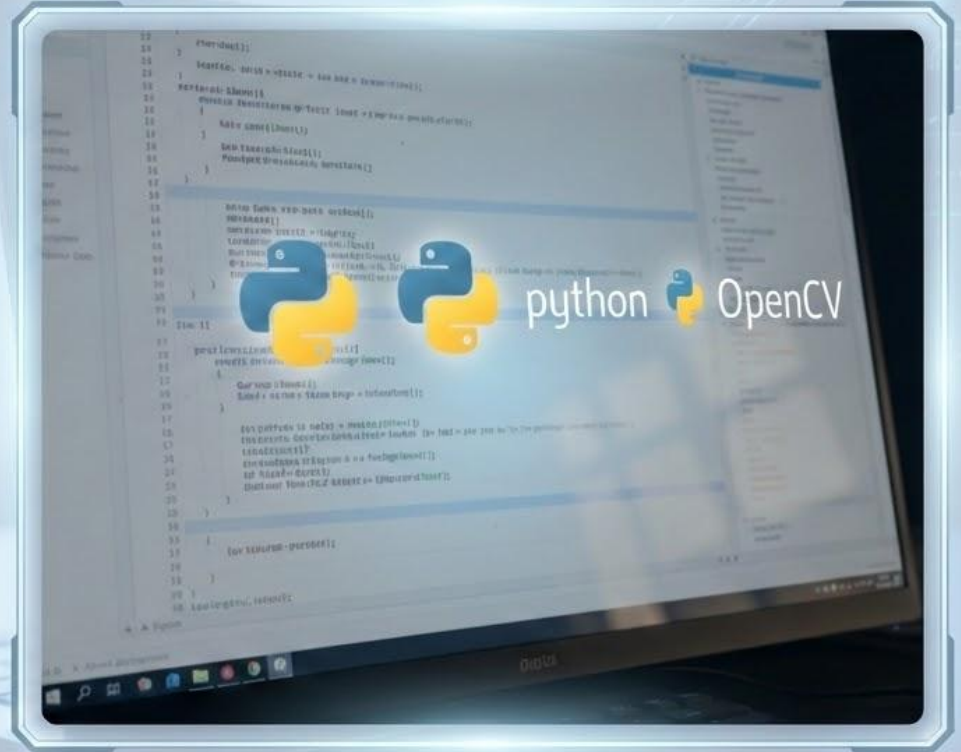
MediaPipe: Face Mesh



NumPy/Pandas: Data processing and structural handling.



Visualization: Matplotlib/Streamlit.

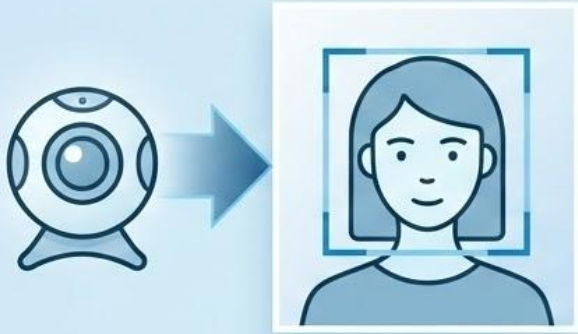


System Architecture



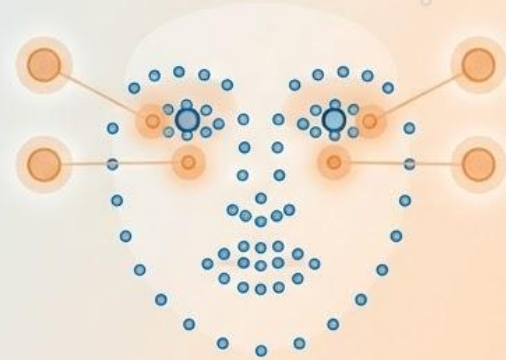
Methodology - Face & Landmark Detection

Detection



Uses Dlib's HOG face detector to locate the face in every frame.

Landmarks



Identifies 68 facial landmarks, specifically tracking 6 key points around each eye.

Methodology - The Core Metric



Metric: Eye Aspect Ratio (EAR)

Logic

High EAR



Eyes Open
(Attentive)

Low EAR



Eyes Closed
(Drowsy)

No Face



Absent

Classification Logic

Status Categories:

Attentive



User is present with
high EAR.

Drowsy



User is present with
EAR below threshold.

Absent



Face is not detected
in the frame.

Project Timeline & Development



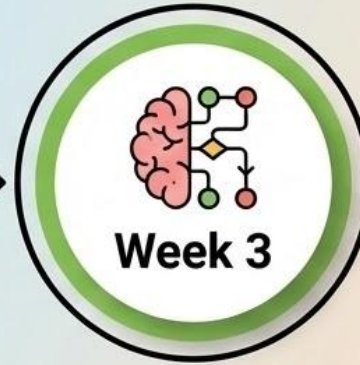
Setup and Object Detection.

Week 1: Focuses on initial system setup, library installation, and implementing the face and eye detection algorithms.



Real-time EAR calculation.

Week 2: Dedicated to developing and refining the real-time Eye Aspect Ratio (EAR) calculation module for drowsiness detection.



Engagement status logic implementation.

Week 3: Involves programming the logic to classify user state (Attentive, Drowsy, Absent) based on EAR and face detection data.



Data logging and cleaning.

Week 4: Concentrates on implementing data logging features for session recording and performing data cleaning and preprocessing.

Performance Metrics

Accuracy



90%

Target performance of 90% accuracy in classification.

Latency



< 1 second

Processing speed of less than one second per frame for real-time performance.

Live Demo: FocusLens

FocusLens | Frame 22



Monitoring Session Complete.

Project Summary: FocusLens



Development Timeline

- **Week 1:** Setup & Detection
- **Week 2:** Real-time EAR
- **Week 3:** Engagement Logic
- **Week 4:** Data Logging



Performance Achievements



Accuracy: 90% Target Met
Latency: < 1 Second
(Real-time)



System Validation

Live Demo Successful
Status: Attentive
Session Monitoring Complete

FocusLens successfully demonstrates a real-time, accurate, and efficient driver drowsiness detection system.